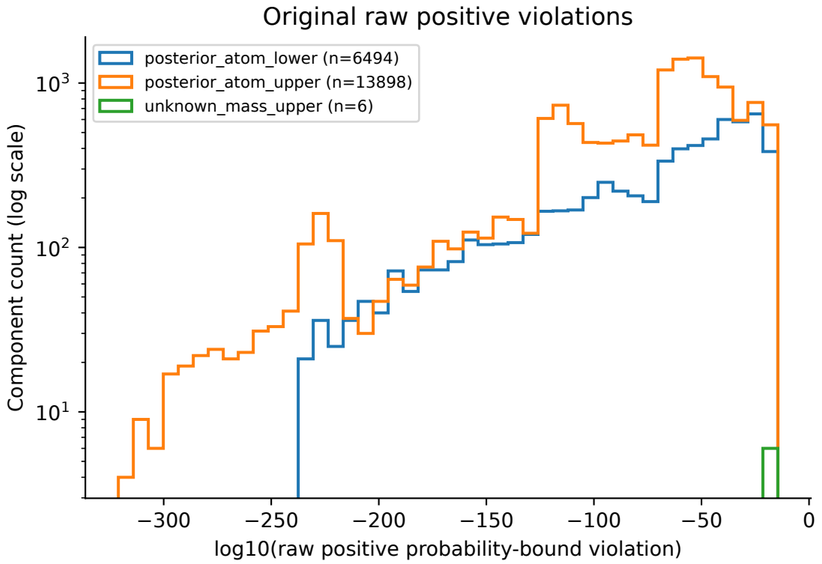
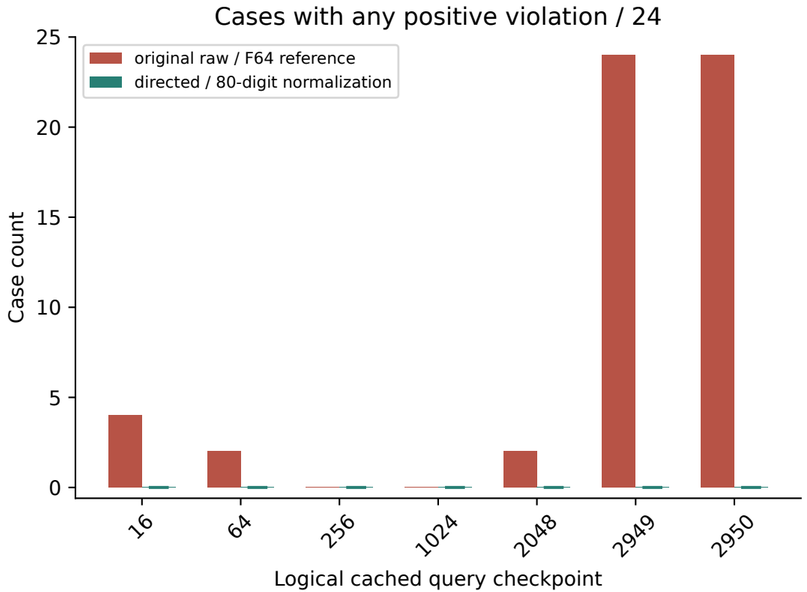
M4 original floating point audit and conditional directed normalization





Normalization audit scope

Status: COMPLETE_RECORDED_CONDITIONAL_AUDIT
Directed bounds are conditional on saved float64 log-potentials.

Original point error versus 80-digit normalization:
6.04e-17 to 5.6e-16
(a normalization diagnostic, not a forward-error bound).

Positive raw log-likelihood components: 0.
No positive raw violation is rounded or clipped away.

Does not certify Maxwell/FEM forward error, formation of the float64 Gaussian potential, or the continuum PDE.

Online use adds certificate cost. Original raw results and their FAIL status stay unchanged.

Counts compare distinct recorded numerical audits: original float64 reference versus independent 80-digit normalization.
These descriptive plots neither replace the original raw gate nor establish Maxwell certification or a publication-level PASS.